Lesson 12 Notes Logic Building Program in Java :

```
int n,r,sum=0;
    System.out.println("Enter any digit:");
    n=so.nextInt();

    while(n!=0)
    {
       r=n%10;
     sum=sum+r;
     n=n/10;
    }
    System.out.println("sum of the digit is:"+sum);
```

n=423 → 9

r=423%10= 3

sum=0+3=3
n=423/10=42

n=42

r=42%10=2
sum=3+2=5
n=42/10=4

n=4

r=4%10=4
sum=5+4=9
n=4/10=0

n=0

```
int n,r,sum=0;
    System.out.println("Enter any digit:");
    n=so.nextInt();

    while(n!=0)
    {
       r=n%10;
     sum=sum+r;
     n=n/10;
    }
    System.out.println("sum of the digit is:"+sum);
```

n=423 → 9

r=423%10= 3

sum=0+3=3
n=423/10=42

n=42

r=42%10=2
sum=3+2=5
n=42/10=4

n=4

r=4%10=4
sum=5+4=9
n=4/10=0

n=0

```
int n,r,rev=0;
    System.out.println("Enter any digit:");
    n=so.nextInt();

    while(n!=0)
    {
      r=n%10;
     rev=(rev*10)+r;
      n=n/10;
    }
    System.out.println("Reverse is:"+rev);
```

$\boxed{n=423}$ → 324

r=423%10=3
rev=(0*10)+3=3
n=423/10=42

$\boxed{n=42}$

r=42%10=2
rev=(3*10)+2=32
n=42/10=4

$\boxed{n=4}$

r=4%10=4
rev=(32*10)+4=324
n=4/10=0

$\boxed{n=0}$

```
int n,r,rev=0;
    System.out.println("Enter any digit:");
    n=so.nextInt();

    while(n!=0)
    {
      r=n%10;
     rev=(rev*10)+r;
      n=n/10;
    }
    System.out.println("Reverse is:"+rev);
```

~~n==rev~~

~~0==121~~

$\boxed{n=423}$ → 324

r=423%10=3
rev=(0*10)+3=3
n=423/10=42

$\boxed{n=42}$

r=42%10=2
rev=(3*10)+2=32
n=42/10=4

$\boxed{n=4}$

r=4%10=4
rev=(32*10)+4=324
n=4/10=0

$\boxed{n=0}$

Palindrome Number

121

121

```
int n,r,rev=0;
    System.out.println("Enter any digit:");
    n=so.nextInt();

    while(n!=0)
    {
      r=n%10;
     rev=(rev*10)+r;
      n=n/10;
    }
    System.out.println("Reverse is:"+rev);
```

~~n==rev~~

~~0==121~~

n=423 → 324

r=423%10=3
rev=(0*10)+3=3
n=423/10=42

n=42

r=42%10=2
rev=(3*10)+2=32
n=42/10=4
n=4

r=4%10=4
rev=(32*10)+4=324
n=4/10=0

n=0

Palindrome Number

121

121

Armstrong number

153

1*1*1=1
5*5*5=125
3*3*3=27

153

Fibonaccie

0
1
1
2
3
5
8
13

```
int n1=0,n2=1,n3;
    System.out.println(n1);//0
    System.out.println(n2);//1

    for(int c=1;c<=10;c++)
    {
        n3=n1+n2;
        System.out.println(n3);//1
        n1=n2;
        n2=n3;
    }
```

n1=0
n2=1
n3=1

n1= 1
n2= 1
n3=1+1=2

n1=1
n2=2
n3= 1+2=3

n1=2
n2=3
n3=2+3=5

Fibonaccie

0
1
1
2
3
5
8
13

```
int n1=0,n2=1,n3;
    System.out.println(n1);//0
    System.out.println(n2);//1

    for(int c=1;c<=10;c++)
    {
        n3=n1+n2;
        System.out.println(n3);//1
        n1=n2;
        n2=n3;
    }
```

n1=0
n2=1
n3=1

n1= 1
n2= 1
n3=1+1=2

n1=1
n2=2
n3= 1+2=3

n1=2
n2=3
n3=2+3=5

while loop
do while loop

```
import java.util.Scanner;

class A

{

  public static void main(String args[])

  {
```

```
        Scanner so=new Scanner(System.in);

        int n,r,sum=0;

        System.out.println("Enter any digit:");

        n=so.nextInt();


        while(n!=0)

        {

           r=n%10;

       sum=sum+r;

       n=n/10;

        }

        System.out.println("sum of the digit is:"+sum);


  }

}
```

```
import java.util.Scanner;

class A

{

  public static void main(String args[])
```

```
 {
        Scanner so=new Scanner(System.in);
        int n,r,rev=0;
        System.out.println("Enter any digit:");
        n=so.nextInt();

        while(n!=0)
        {
           r=n%10;
       rev=(rev*10)+r;
        n=n/10;
        }
        System.out.println("Reverse is:"+rev);

 }
}
```

```
import java.util.Scanner;
class A
{
```

```
public static void main(String args[])
{
    Scanner so=new Scanner(System.in);
    int n,r,rev=0;
    System.out.println("Enter any digit:");
    n=so.nextInt();
    int x=n;
    while(n!=0)
    {
        r=n%10;
    rev=(rev*10)+r;
    n=n/10;
    }
    System.out.println("Reverse is:"+rev);


    if(x==rev)
    {
    System.out.println("Given number is a palindrome
number:");
    }
```

```
        else
        {
        System.out.println("Given number is not palindrome
number:");
        }

  }
}
```

```
class A
{
  public static void main(String args[])
  {
        int n1=0,n2=1,n3;
        System.out.println(n1);//0
        System.out.println(n2);//1

        for(int c=1;c<=10;c++)
        {
```

```
            n3=n1+n2;

            System.out.println(n3);//1

            n1=n2;

            n2=n3;

        }

  }

}
```